



Picturing the Floor Plan

Mental-Imagery Vividness Predicts Who Lingers at the Building Directory, Not Signage Clarity

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Introduction

Wayfinding audits treat a lingering visitor as a signage failure. Converting a floor plan into a mental route takes people different amounts of time — is dwell time measuring confusion, or a stable difference in how vividly people picture space?

Research questions

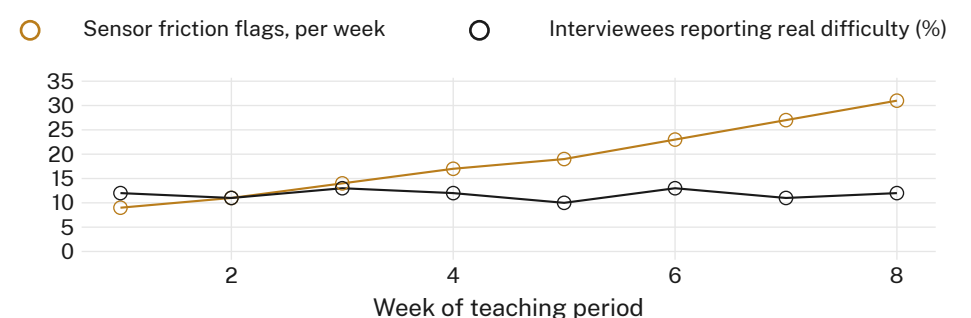
- Does self-reported imagery vividness predict directory-board dwell time?
- Does the sensor's friction flag track genuine wayfinding difficulty?
- Do flagged buildings differ in imagery-vividness mix, not signage quality?

Approach

- n = 166 directory-board pauses intercepted and interviewed · 9 buildings
- one teaching period
- VVIQ imagery-vividness self-report plus a 2-minute route-recall task
- Sensor's existing 25-second threshold held unchanged; no recalibration attempted

What we found

Imagery-vividness quartile predicted dwell time more strongly than any building-level signage variable measured. Sensor-flagged buildings reported no more wayfinding difficulty than unflagged ones.



Sensor-issued wayfinding-friction flags rose from 9 to 31 per week across weeks 1–8 of Semester 1, 2026 (n = 166 board pauses); interviewees reporting genuine navigation difficulty held between 10% and 13% throughout.

Discussion

The sensor's threshold is tracking the imagery-vividness mix of who happens to pass, not the board's clarity. A signage redesign aimed at the flag would edit a wall that was never the problem, and the redesign brief now in circulation cites the flag count as its evidence base without once naming the cohort it actually describes. Next: validate the threshold against VVIQ composition before it authorises any redesign spend.

Further reading

1. Marks, D. F. (1973). Visual imagery differences in the recall of pictures. *British Journal of Psychology*, 64(1), 17–24. doi:10.1111/j.2044-8295.1973.tb01322.x
2. Zeman, A., Dewar, M., & Della Sala, S. (2015). Lives without imagery — congenital aphantasia. *Cortex*, 73, 378–380. doi:10.1016/j.cortex.2015.05.019
3. Keogh, R., & Pearson, J. (2018). The blind mind: no sensory visual imagery in aphantasia. *Cortex*, 105, 53–60. doi:10.1016/j.cortex.2017.10.012
4. Weisberg, S. M., & Newcombe, N. S. (2016). How do (some) people make a cognitive map? Routes, places, and working memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 42(5), 768–785. doi:10.1037/xlm0000200
5. Ishikawa, T., & Montello, D. R. (2006). Spatial knowledge acquisition from direct experience in the environment. *Cognitive Psychology*, 52(2), 93–129. doi:10.1016/j.cogpsych.2005.08.003
6. Hegarty, M., Richardson, A. E., Montello, D. R., Lovelace, K., & Subbiah, I. (2002). Development of a self-report measure of environmental spatial ability. *Intelligence*, 30(5), 425–447. doi:10.1016/S0160-2896(02)00116-2

Human ethics approval 2025/118; VVIQ responses de-identified before matching to sensor logs; thresholds pre-registered.

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